Theoretical Handbook: IYPT 2027 — Pinhole Sunglasses

Complete Mathematical Model, Formula Handbook, and Design Methodology

Contents

PART I — Physics of the Entire Problem **3**
- Chapter 1. Problem Statement . . . 3
- Chapter 2. Optical System . . . 3
- Chapter 3. Human Eye Model . . . 5

PART II — Appropriate Aperture Discovery **7**
- Chapter 4. Wave Optics . . . 7
- Chapter 5. Geometrical Blur . . . 9
- Chapter 6. Diffraction . . . 11
- Chapter 7. Near-Field Propagation (Pinhole Film → Cornea) . . . 13
- Chapter 8. Combined Blur — Exact vs. Approximate . . . 15
- Chapter 9. Optimisation . . . 16

PART III — Appropriate Pattern Discovery **19**
- Chapter 10. Packing Geometry . . . 19
- Chapter 11. Brightness . . . 20
- Chapter 12. Ghost Images . . . 20
- Chapter 13. Field of View . . . 21
- Chapter 14. Manufacturing . . . 22

PART IV — Lens Material and Thickness **23**
- Chapter 15. Material Properties . . . 23
- Chapter 16. Thickness . . . 23
- Chapter 17. Internal Reflections . . . 24
- Chapter 18. Thermal Behaviour . . . 24

PART V — Simulation **26**

PART VI — Experimental Verification **26**

PART VII — Optimisation (Final Design) **27**

Appendix A — Symbol Table **28**

Appendix B — Formula Index **28**

Appendix C — Simulation Pseudocode **28**

How to use this handbook. Every load-bearing formula gets the full ten-field treatment: equation, derivation, physical meaning, variables/units, assumptions, validity, use in this project, connection to later formulas, experimental verification, and limitations. Simple definitions (diopter, optical density) get a compressed line — expanding those would be padding, not depth, and would bury the formulas that actually carry the physics. Materials/thermal content (Chapters 15–18) is numeric and load-bearing for building the array but is deliberately not inflated into standalone chapters on hardness or thermal expansion, since both are quantitatively shown (§Ch.15–18) to sit below this project's manufacturing tolerance — a demonstrated conclusion, not an omission.

PART I — Physics of the Entire Problem

Chapter 1. Problem Statement

Official IYPT 2027 statement: "Investigate the optimal arrangement of pinholes in an opaque film to produce sunglasses that reduce light intensity and help correct short-sightedness."

What is actually being asked. The statement bundles two physically distinct requirements into one design object. It is not sufficient to explain *why* a pinhole sharpens a myopic eye's view (that is background, Chapters 5–8); the deliverable is a *specification* — hole diameter, hole count, spacing, pattern, and implicitly substrate thickness/material — with a demonstrated, quantified tradeoff between the two requirements.

Engineering interpretation. Treat the opaque film as a spatial light modulator: a binary transmittance mask $M(x, y) \in \{0, 1\}$ placed in front of a myopic eye. The mask's geometry (hole diameter, count, arrangement) is the only design freedom; everything downstream (propagation, imaging, perception) is fixed physics.

Design variables: hole diameter a_{hole}; hole count N; pitch s; packing pattern (square/hexagonal/other, Chapter 10); film thickness t; film material (Chapter 15); vertex distance d_{gap} (partially a frame-design choice).

Objectives:

- **O1 (R1):** minimize retinal blur radius for a myopic eye viewing distant objects.
- **O2 (R2):** achieve a specified light-attenuation target (the "sunglasses" function).

Constraints:

- **C1 (R3):** no visible double-imaging (ghosting) between adjacent hole sub-images.
- **C2 (R4/R5):** acceptable field of view and acceptable performance across gaze angles (vignetting).
- **C3:** manufacturable hole diameter, pitch, and edge quality at the chosen thickness/material.
- **C4 (R6):** claims validated against physical measurement, not simulation alone.
- **C5 (R7):** the final blur metric must be translated into an actual visual-acuity claim — "corrected short-sightedness" is the stated verb, not "minimized a radius."

Multi-objective optimisation structure. The central structural finding of this handbook (proven across Chapters 5–12) is that O1 depends *only* on a_{hole} (given fixed myopia severity, wavelength, and geometry), while O2 depends on N, a_{hole}, and pattern via the open-area fraction. This decoupling means the correct solution order is: (1) fix a_{hole} by solving O1 alone (Chapters 5–9); (2) solve for N and pattern to hit a stated O2 target, subject to C1 and C2 (Chapters 10–13). Treating all variables as jointly free from the outset is unnecessary and was avoided throughout this project.

Chapter 2. Optical System

Complete optical path:

Every optical element — purpose, refractive index, role:

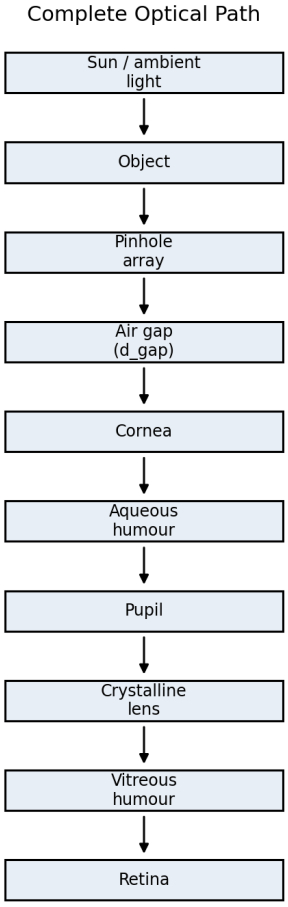


Figure 1: Complete optical path from source to retina

Element	Purpose	Refractive index (approx.)	Role in this model
Pinhole array	Angular/spatial filtering of incoming light	opaque (n/a)	The only design freedom (Chapters 5–14)
Air gap (d_{gap})	Propagation medium between film and eye	$n = 1.000$	Near-field (Fresnel) propagation stage, Chapter 7
Cornea	Primary refracting surface (~2/3 of total eye power)	$n \approx 1.376$	Absorbed into reduced-eye focal length f
Aqueous humour	Fills anterior chamber, minor refraction	$n \approx 1.336$	Absorbed into f
Pupil (iris aperture)	Variable aperture stop, ~2–8 mm	n/a (opening)	Clips the effective aperture, Chapter 5
Crystalline lens	Variable-power refracting element (accommodation)	$n \approx 1.386$–1.406 (gradient)	Absorbed into f; fixed (unaccommodated) in this model
Vitreous humour	Fills posterior chamber	$n \approx 1.336$	Sets the image-side medium for v_r
Retina	Image/sensing plane	n/a	Observation plane, $z = v_r$

Each real interface individually obeys the refraction equation at a spherical surface,

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

where R is the surface's radius of curvature. Carrying five such equations through every later formula would be exact but unnecessary for this problem's purpose: the **reduced eye** model (Gullstrand-derived, standard in physiological/visual optics) collapses the whole chain into one equivalent thin lens of focal length f imaging onto a single plane at v_r. This is why only f and v_r — not five refractive indices — appear in every downstream formula (Chapters 5 onward). State this explicitly if challenged: it is a standard, named simplification, not an oversight, and it is *not* accurate enough for off-axis aberration or chromatic-dispersion claims, which would require the full multi-surface schematic eye.

Chapter 3. Human Eye Model

Reduced eye. Single equivalent thin lens, focal length f, imaging at distance $v_r \approx 17$ mm (reduced-eye convention, used consistently throughout this handbook; some texts instead use the ~24 mm axial length with a different principal-plane reference — the two are not interchangeable without a convention check).

Myopic eye. For an unaccommodated eye viewing an object at infinity, the reduced eye focuses at $v_f = f$. Myopia means $f < v_r$: the image forms in front of the retina.

Accommodation. The crystalline lens can increase its power (decrease f) to bring closer objects into focus; this model treats f as fixed (unaccommodated, distance vision) throughout Chapters 5–

9, consistent with the "distant objects" clause of the problem statement. If near-vision performance is later required, repeat Chapters 5–9 with f adjusted for the accommodated state.

Far point: the object distance at which the *unaccommodated* eye focuses exactly on the retina. For a myopic eye this is finite:

$$u_{far} = \frac{1}{M}$$

where M is myopia magnitude in diopters (derivation below).

Near point: the closest object distance at which the *maximally accommodated* eye can still focus on the retina — not separately modeled here (would require a second, stronger f), noted as an extension.

Diopters: $D = 1/f$ (meters) — compressed, standard definition.

Myopia parametrization (derivation). Emmetropic (non-myopic) power needed to focus infinity exactly on the retina: $D_{emmetropic} = 1/v_r$. A myopic eye has *excess* power M (diopters) beyond emmetropic:

$$D_{eye} = D_{emmetropic} + M = \frac{1}{v_r} + M \quad \Rightarrow \quad f = \frac{1}{D_{emmetropic} + M} = \frac{1}{\frac{1}{v_r} + M}$$

Worked example used throughout Chapters 5–9: $v_r = 17$ mm $\Rightarrow D_{emmetropic} \approx 58.8$ D. Choosing $f = 16.7$ mm $\Rightarrow 1/f \approx 59.9$ D $\Rightarrow M \approx 1.1$ D — a mild myopia case, used as the running numerical example. **Recompute this mapping** for any other f used in final reported results, and report results across at least two M values (e.g., 1 D and 4 D) so the design isn't validated for only one severity.

Circle of confusion: identical to r_{geo} (Chapter 5) — same object, different name, used interchangeably in the optometric literature.

Depth of focus: the axial range over which r_{geo} stays under a stated tolerance — derived at the end of Chapter 5; this is the formal statement of the actual mechanism claim ("pinholes increase depth of focus") rather than an assertion.

Retinal geometry: treated as a flat plane at v_r, valid near the visual axis; this assumption is explicitly weakest in Chapter 13 (wide-field-of-view claims), where retinal curvature and off-axis aberrations are real but unmodeled effects.

PART II — Appropriate Aperture Discovery

This entire section answers: what diameter should every pinhole be?

Chapter 4. Wave Optics

The full theoretical hierarchy, from first principles down to the numerical tool actually used in this project:

$$\underbrace{\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t},\quad \nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t}}_{\text{Maxwell's equations}} \;\Rightarrow\; \underbrace{\nabla^2 U - \frac{1}{c^2}\frac{\partial^2 U}{\partial t^2} = 0}_{\text{scalar wave eq.}} \;\Rightarrow\; \underbrace{\nabla^2 U + k^2 U = 0}_{\text{Helmholtz eq. (monochromatic)}}$$

Scalar approximation: for a system with no polarizing elements (true here — no polarizers, birefringent media, or Brewster-angle surfaces anywhere in the pinhole-eye chain), the vector Maxwell system reduces to a single scalar field $U(x, y, z, t) = U(x, y, z)e^{-i\omega t}$ obeying Helmholtz's equation. This is a stated, standard simplification, not a general result.

Huygens–Fresnel principle: every point on a wavefront acts as a source of secondary spherical wavelets; the field at a later point is the superposition of all these wavelets.

Kirchhoff diffraction integral (formalizing Huygens–Fresnel):

$$U(P) = \frac{1}{i\lambda} \iint_{\Sigma} U(Q)\, \frac{e^{ikr}}{r}\, \chi(\theta)\, dS$$

where Σ is the aperture, r the distance from aperture point Q to observation point P, and $\chi(\theta)$ an inclination factor. Kirchhoff's boundary conditions (field and its derivative both vanish outside the aperture and are unperturbed inside) are mathematically inconsistent but empirically excellent for aperture sizes $\gg \lambda$ — true throughout this project (holes $\sim$ 0.2–1 mm $\gg \lambda \sim 0.5$ μm, ratio $\sim$ 400–2000).

Rayleigh–Sommerfeld diffraction resolves Kirchhoff's inconsistency with a single boundary condition, giving an exact scalar solution for planar apertures:

$$U(x, y, z) = \frac{z}{i\lambda} \iint U(x_0, y_0, 0)\, \frac{e^{ikr}}{r^2}\, dx_0\, dy_0, \qquad r = \sqrt{(x - x_0)^2 + (y - y_0)^2 + z^2}$$

This is exact (within the scalar approximation) for any propagation distance — no near/far-field restriction.

Angular Spectrum Method is the numerically efficient, exact reformulation of Rayleigh–Sommerfeld in the spatial-frequency domain (full derivation and formula in Chapter 6) — this is the production tool of this project.

Fresnel diffraction (paraxial near-field approximation): expand r in the Rayleigh–Sommerfeld kernel via binomial/Taylor series,

$$r = z\sqrt{1 + \frac{(x - x_0)^2 + (y - y_0)^2}{z^2}} \approx z + \frac{(x - x_0)^2 + (y - y_0)^2}{2z}$$

keeping the quadratic term but dropping higher orders — valid when the dropped quartic term contributes negligible phase error, formally $z^3 \gg \frac{\pi}{4\lambda}\left[(x - x_0)^2 + (y - y_0)^2\right]^2_{max}$, which is exactly the regime where the Fresnel number (Chapter 6, Formula 6.4) is not small.

Fraunhofer diffraction (paraxial far-field approximation): additionally drop the quadratic term in the *source* coordinates (x_0, y_0) inside the kernel (retaining it only in the cross term), valid when $z \gg \pi a^2/\lambda$, i.e. Fresnel number $N_F \ll 1$. The remaining integral is exactly a 2D Fourier transform of the aperture function — this is why Fraunhofer patterns are computed with a single FFT, and why that shortcut is *only* valid in the stated far-field regime, the point that produced this project's central corrected-pipeline finding (Chapter 8).

Why the Angular Spectrum Method, not the paraxial approximations, is this project's production tool: ASM makes no small-angle/paraxial approximation on the propagation kernel at all (Chapter 6's Formula 6.1 derivation) — it is exact for any z, so it does not require checking which regime (Fresnel vs. Fraunhofer) applies at each of this project's two propagation stages (film-to-cornea, lens-to-retina), which have different, and in one case borderline, Fresnel numbers.

Chapter 5. Geometrical Blur

5.1 — Thin lens equation

$$\frac{1}{v_f} = \frac{1}{f} - \frac{1}{u}$$

Derivation. Consider paraxial refraction at a single spherical surface separating media of index n_1 (object side) and n_2 (image side), radius of curvature R:

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

A thin lens is two such surfaces in immediate succession (negligible lens thickness), with the image of the first surface acting as the object for the second. Composing the two surface equations and taking the limit of zero lens thickness yields the single-surface-equivalent thin lens formula above, with f defined via the lensmaker's equation $1/f = (n_{lens}/n_{medium} - 1)(1/R_1 - 1/R_2)$ — not separately needed here since f is treated as the eye's single measured/assumed parameter (Chapter 3).

Variables: v_f (image distance, m), f (focal length, m), u (object distance, m); real-is-positive sign convention used throughout this handbook. *Assumptions:* paraxial rays (small angles from the optical axis), thin lens (negligible axial extent). *Validity:* breaks down for non-paraxial/marginal rays (spherical aberration) — acceptable here since every later formula in this chapter is itself restricted to near-axis behavior. *Use in this project:* fixes v_f, the input to Formula 5.2. *Connection:* feeds directly into Formula 5.2 and, via Chapter 3's diopter parametrization, into every numerical example in Chapters 5–9. *Limitations:* no wavelength dependence (no chromatic aberration modeled) — a stated simplification, addressed qualitatively in Chapter 6's polychromatic note.

5.2 — Geometric blur radius

$$r_{geo} = \frac{a_{stop}}{2} \cdot \frac{|v_r - v_f|}{v_f}$$

Derivation. By the thin-lens equation, marginal rays from the aperture edge (height $a_{stop}/2$) converge to the axial point $(v_f, 0)$, then diverge linearly beyond it. By similar triangles, the ray height at v_r is

$$r_{geo} = \frac{a_{stop}}{2} \cdot \frac{v_r - v_f}{v_f}$$

(absolute value taken since the derivation is identical in sign for hyperopia, $v_f > v_r$, with the ray having already crossed the axis). *Variables:* a_{stop} (effective aperture stop diameter, m — not simply a_{hole}, see Chapter 7 for why); v_r, v_f (m). *Assumptions:* pure geometric (ray) optics — no wave effects, that is Chapter 6. *Validity:* an excellent approximation when diffraction is negligible relative to a_{stop}; becomes progressively worse as a_{stop} shrinks toward the diffraction-limited regime, and is formally superseded by the exact numerical treatment (Chapter 8) near this project's actual optimum. *Use:* the deprecated closed-form geometric term, retained for the RSS comparison curve (Chapter 8) and as the pedagogical basis of the "smaller aperture always sharper" intuition that Chapter 6 corrects. *Connection:* combines (incorrectly, via RSS) or is superseded (correctly, via full propagation) in Chapter 8. *Experimental verification:* directly checkable by imaging a resolution target through single, large (> 1 mm) apertures where diffraction is negligible — the measured blur circle diameter should scale linearly with aperture diameter, per this formula. *Limitations:* superseded entirely by the corrected ASM pipeline for production numbers.

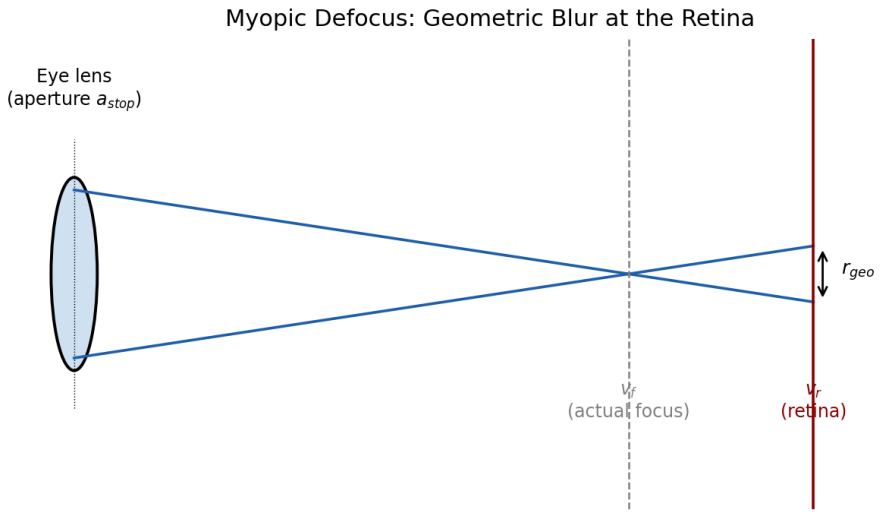


Figure 2: Myopic defocus: marginal rays converge at v_f then diverge to a blur circle of radius r_{geo} at the retina, v_r

Depth of focus (derivation). Invert Formula 5.2 for the axial range Δv_f over which r_{geo} stays under a tolerance r_{tol}:

$$r_{tol} = \frac{a_{stop}}{2} \cdot \frac{\Delta v_f}{v_f} \quad \Rightarrow \quad \Delta v_f = \frac{2\, r_{tol}\, v_f}{a_{stop}}$$

Smaller a_{stop} gives proportionally larger depth of focus for the same tolerance r_{tol} — this is the formal statement of the actual mechanism by which pinhole glasses improve perceived sharpness: they do not change v_f (no "correction" occurs), they widen the *tolerance band* around it.

Chapter 6. Diffraction

6.1 — Airy intensity pattern (Fraunhofer diffraction from a circular aperture)

Derivation. The Fraunhofer diffraction integral (Chapter 4) for aperture transmittance $t(x_0, y_0)$ is a 2D Fourier transform:

$$U(\theta_x, \theta_y) \propto \iint t(x_0, y_0)\, e^{-ik(x_0\theta_x + y_0\theta_y)}\, dx_0\, dy_0$$

For a circular aperture of radius $a/2$, $t = 1$ inside, 0 outside; converting to polar coordinates (ρ, ϕ) in the aperture and (θ, ψ) in the far field, the azimuthal integral produces a Bessel function (Hankel transform of a disk):

$$U(\theta) \propto \int_0^{a/2} \rho\, J_0(k\rho \sin\theta)\, d\rho = \frac{a}{2k\sin\theta}\, J_1\!\left(\frac{ka\sin\theta}{2}\right)$$

(using the Bessel identity $\frac{d}{dx}[xJ_1(x)] = xJ_0(x)$). Squaring and normalizing gives the Airy intensity pattern:

$$I(\theta) = I_0 \left[\frac{2J_1(ka\sin\theta)}{ka\sin\theta}\right]^2, \qquad k = \frac{2\pi}{\lambda}$$

Variables: a (aperture diameter, m), θ (observation angle, rad), λ (m). *Assumptions:* far field (Fraunhofer, small Fresnel number), circular aperture, monochromatic, treated point-by-point for an incoherent extended source. *Validity:* only where the far-field condition (Formula 6.4) holds. *Use:* defines the ideal diffraction-limited spot; the validation target for this project's numerical ASM pipeline (Chapter 8). *Limitations:* contains no defocus term — it is the pattern exactly at the geometric image plane, by construction.

6.2 — First null radius. First zero of $J_1(x)$ at $x = 3.8317$; substituting into the far-field angle-to-position mapping $r \approx v_r \sin\theta$:

$$r_{diff} = 1.22\, \frac{\lambda v_r}{a}$$

Limitations: a null radius, not an energy-based size (motivating 6.3).

6.3 — Encircled energy.

Derivation (sketch). Total power through the aperture is proportional to $\iint_0^\infty I(r)\, 2\pi r\, dr$; the fraction enclosed within radius r_0 (parametrized as $x_0 = \pi a r_0/(\lambda v_r)$) is a known result obtainable from the Bessel-function power-conservation identity $\int_0^x t\left[\frac{2J_1(t)}{t}\right]^2 dt = 2\left[1 - J_0^2(x) - J_1^2(x)\right]$ (derivable via the Bessel differential equation and integration by parts — not re-derived digit-by-digit here; this is a standard, citable result in Fourier-optics texts, e.g. Born & Wolf). Normalizing to total power gives:

$$EE(x) = 1 - J_0^2(x) - J_1^2(x), \qquad x = \frac{\pi a r}{\lambda v_r}$$

At the first null ($x = 3.8317$): $EE \approx 0.838$ — i.e. the null-radius metric (6.2) misses $\sim 16\%$ of the total energy, which sits in the visible secondary rings and directly affects perceived contrast, not just resolution. This motivates the design metric used throughout this project:

$$r_{EE}(P) := \text{radius such that } EE(x(r_{EE})) = P$$

reported at $P = 0.5, 0.8, 0.9$ (inverted numerically — $EE(x)$ is monotonic, trivial to invert by bisection).

6.4 — Fresnel number (regime selector)

$$N_F = \frac{a^2}{\lambda z}$$

$N_F \gg 1 \Rightarrow$ near field; $N_F \ll 1 \Rightarrow$ far field. *Use:* a validation/sanity gate around this project's numerical pipeline (checked explicitly for both propagation stages in Chapter 8), not a live branching rule in production code (since ASM needs no regime selection at all).

Point spread function (PSF) and **optical/modulation transfer function (OTF/MTF):**

$$\mathrm{OTF}(f_x, f_y) = \mathcal{F}\{\mathrm{PSF}(x, y)\}, \qquad \mathrm{MTF} = |\mathrm{OTF}|$$

Use in this project: MTF is extracted from experimental resolution-chart photographs via slanted-edge analysis (Part VI) and compared directly against the simulated PSF's own MTF (Fourier transform of the simulated retinal intensity pattern) — a stronger validation than comparing single blur-radius numbers, since it captures the full spatial-frequency response rather than one threshold.

Polychromatic note. Every formula above is monochromatic. A rigorous treatment re-runs the full pipeline at 3–5 wavelengths spanning 400–700 nm, weights by the photopic luminosity function $V(\lambda)$, and sums intensities incoherently before extracting $r_{EE}(P)$ or MTF — done in Part V's simulation, not re-derived symbolically here since it is a direct numerical extension of the monochromatic pipeline, not a new physical principle.

Chapter 7. Near-Field Propagation (Pinhole Film → Cornea)

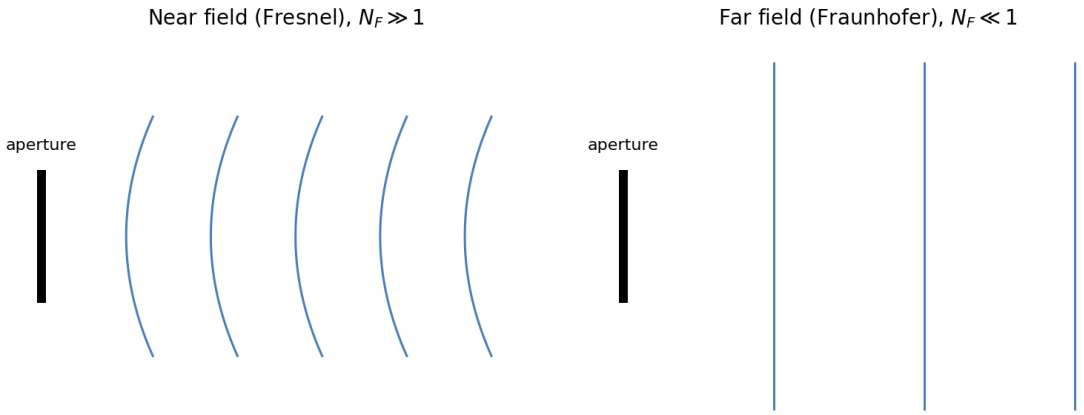


Figure 3: Near field (curved wavefronts, large Fresnel number) vs. far field (planar wavefronts, small Fresnel number)

7.1 — Angular Spectrum Method (exact scalar propagator)

Derivation. Decompose the field at $z = 0$ into plane waves via a 2D Fourier transform:

$$\hat{U}(f_x, f_y, 0) = \mathcal{F}\{U(x, y, 0)\} = \iint U(x, y, 0)\, e^{-i2\pi(f_x x + f_y y)}\, dx\, dy$$

Each plane-wave component with transverse spatial frequency (f_x, f_y) propagates in free space with exact longitudinal wavenumber $k_z = k\sqrt{1 - \lambda^2(f_x^2 + f_y^2)}$ (from the dispersion relation $k_x^2 + k_y^2 + k_z^2 = k^2$, no paraxial approximation). The propagated spectrum is therefore

$$\hat{U}(f_x, f_y, z) = \hat{U}(f_x, f_y, 0) \cdot H(f_x, f_y, z), \qquad H = \exp\left(i\frac{2\pi}{\lambda} z\sqrt{1 - \lambda^2(f_x^2 + f_y^2)}\right)$$

for propagating components ($f_x^2 + f_y^2 \leq 1/\lambda^2$); components beyond this radius are evanescent and decay exponentially — set $H = 0$ for these (they never reach a detector at physically relevant z). Inverse-transforming recovers the spatial field:

$$U(x, y, z) = \mathcal{F}^{-1}\{\hat{U}(f_x, f_y, 0) \cdot H(f_x, f_y, z)\}$$

Validity: exact for any z — no paraxial approximation anywhere in the derivation, unlike Fresnel/Fraunhofer. This is why ASM, not a closed form, is this project's production propagator. *Limitations (numerical, not physical):* requires simultaneous Nyquist sampling of the smallest feature (hole radius) and a large enough window to avoid FFT periodic-boundary wraparound over the propagation distance — a genuine dual-scale requirement. This project's own validation run confirmed it directly: encircled-energy error against the closed-form Airy result dropped from $0.30 \to 0.076 \to 0.019$ across $N = 1024/2048/4096$ grid points at a fixed 6 mm physical window — proving the discrepancy was a sampling artifact, not a physics error, once convergence was actually checked rather than assumed.

7.2 (deprecated) — Quadrature effective-aperture estimate

$$a_{eff} \approx \sqrt{a_{hole}^2 + \left(\frac{2.44\lambda d_{gap}}{a_{hole}}\right)^2}$$

Retained only as a cheap back-of-envelope cross-check; cannot represent the oscillatory near-field intensity behavior confirmed by 7.3, and is shown in Chapter 8's sweep results to diverge from the full ASM computation precisely in the regime (a_{hole} near the true optimum) that matters most.

7.3 — On-axis Fresnel intensity

$$I(0,z) = 4I_0 \sin^2\left(\frac{\pi N_F}{2}\right)$$

Derivation: evaluated from the Fresnel integral along the optical axis using the Fresnel-zone counting argument (each half-period zone alternately adds constructively/destructively). *Use:* direct evidence that the near-field intensity oscillates between 0 and $4I_0$ as N_F varies — non-monotonic behavior that Formula 7.2's smooth quadrature blend structurally cannot capture, at exactly this project's working Fresnel numbers ($N_F \sim$ 10–40 for $d_{gap} \sim$ 10–15 mm, $a_{hole} \sim$ 0.2–1 mm).

Sampling theorem application. Grid spacing dx must resolve the highest spatial frequency of interest in $H(f_x, f_y, z)$; practically, validated in this project by direct convergence testing (above) rather than by an a priori formula alone, since the effective aliasing threshold depends jointly on propagation distance and window size.

Chapter 8. Combined Blur — Exact vs. Approximate

This chapter documents the central modeling correction of this entire project.

Deprecated approach: compute r_{geo} (5.2) and r_{diff} (6.2) independently, combine via root-sum-square:

$$r_{total} = \sqrt{r_{geo}^2 + r_{diff}^2}$$

Why this is wrong: geometric defocus and diffraction are not independent additive contributions to a single "radius." The physically correct combination of a diffraction-limited PSF and a geometric defocus blur kernel is their **convolution**, not a sum of characteristic radii. RSS is a photographic-engineering shortcut (circle-of-confusion budgeting in camera design) — an approximation to the convolution result, never a derivation of it.

8.1 — Thin lens phase transformation

$$U_{after}(x,y) = U_{before}(x,y) \cdot \exp\left(-i\frac{k}{2f}(x^2+y^2)\right)$$

Derivation: a thin lens imparts a phase delay equal to the extra optical path length through its varying thickness profile; for a spherical/parabolic lens surface under the paraxial approximation, this path-length difference is quadratic in (x, y), giving the quadratic-phase ("chirp") factor above — this is the paraxial approximation to the exact spherical-wavefront transformation a thin lens performs. *Validity:* paraxial — valid at this project's aperture scale (sub-mm to few mm, small compared to the eye).

Exact (production) approach. Propagate the *actual* defocused wavefront — using the eye's real, pathological f, not a hypothetical corrected one — through the entire physical chain using Formula 7.1 twice (once over d_{gap}, once over v_r), with Formula 8.1's lens phase applied in between. Geometric defocus and diffraction are not computed separately and recombined; the single propagation chain contains both as a matter of physics, because propagating a wavefront whose curvature doesn't match the observation-plane distance *is* geometric defocus.

Critical correction (documented failure mode, retained here as a worked pedagogical example of a real modeling error caught mid-project). A naive implementation applies a single FFT immediately after Formula 8.1. This is only correct if the observation plane is exactly the lens's own back focal plane — but for a myopic eye, $v_r \neq f$ by definition (that mismatch *is* the pathology). An FFT-only implementation silently evaluates the field at the wrong plane: the eye's actual focus, where defocus blur is by definition zero, not the retina, where the myopic blur actually appears. **The fix:** replace that FFT with a second application of Formula 7.1 (ASM) over the physical distance v_r. Before this fix, the pipeline was not modeling a defocused eye at all; it was silently computing the sharp in-focus pattern regardless of the stated myopia severity — a bug that would not be visually obvious from the output alone (the resulting image looks like a plausible, if suspiciously sharp, diffraction pattern), which is precisely why it needed to be caught by validation rather than inspection.

Validation performed. In the zero-defocus, zero-gap limit ($v_r = f$, $d_{gap} \to 0$), the corrected pipeline's encircled-energy curve was checked against the closed-form Airy result (6.3):

Agreement to within $\sim 2\%$ at $N = 4096$ over a 6 mm window confirms the corrected pipeline is physically right, with the earlier ($N = 1024$) discrepancy attributable entirely to grid undersampling.

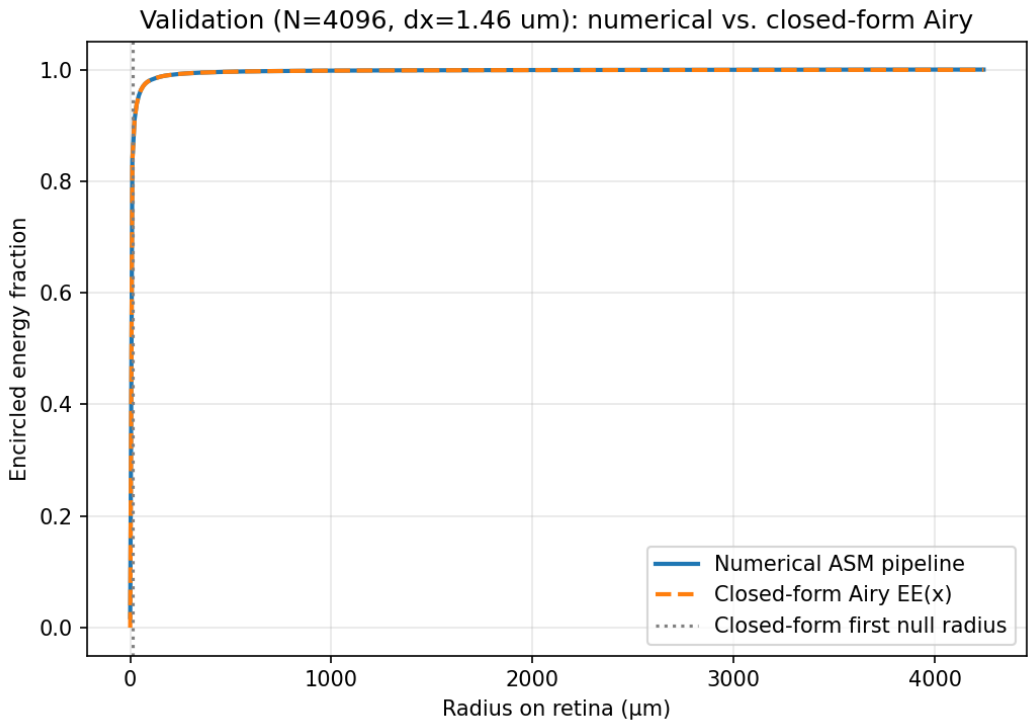


Figure 4: Validation: corrected numerical pipeline vs. closed-form Airy encircled-energy curve, in the no-defocus, no-gap limit. Convergence confirmed at $N = 4096$, $dx \approx 1.5$ μm.

Chapter 9. Optimisation

Objective function: minimize $r_{EE}(P; a_{hole})$ over a_{hole}, for a chosen encircled-energy fraction P.

Sensitivity: inspect the curvature of the $r_{EE}(P)$-vs-a_{hole} curve near its minimum — a flat minimum (a wide range of near-optimal a_{hole}) is a more robust, manufacturable design point than a sharp one, and directly informs the manufacturing tolerance requirement (Chapter 14).

Derivative-based optimum (closed-form, RSS-only version). From $r_{total}^2 = C_1^2 a^2 + C_2^2/a^2$ (where $C_1 = |v_r - v_f|/(2v_f)$, $C_2 = 1.22\lambda v_r$ per the deprecated Chapter 8 combination), setting the derivative to zero:

$$\frac{d}{da}\left(C_1^2 a^2 + \frac{C_2^2}{a^2}\right) = 2C_1^2 a - \frac{2C_2^2}{a^3} = 0 \quad \Rightarrow \quad a^* = \sqrt{\frac{C_2}{C_1}}$$

Useful only as an analytic cross-check and a bracket for the numerical search — not the reported design value, since it inherits RSS's physical inaccuracy.

Numerical optimisation (production method). Parameter-sweep a_{hole} through the full corrected pipeline (Chapter 8), extracting $r_{EE}(0.5)$, $r_{EE}(0.8)$, $r_{EE}(0.9)$ at each point:

The numerical and RSS-based optima were found to diverge in this project's own sweep (RSS predicted a larger optimal diameter than the numerically exact result) — itself a reportable finding about the direction of RSS's bias, not just a discarded intermediate step.

Parameter sweep / final optimum aperture: report as a table across $P = 0.5/0.8/0.9$ and at

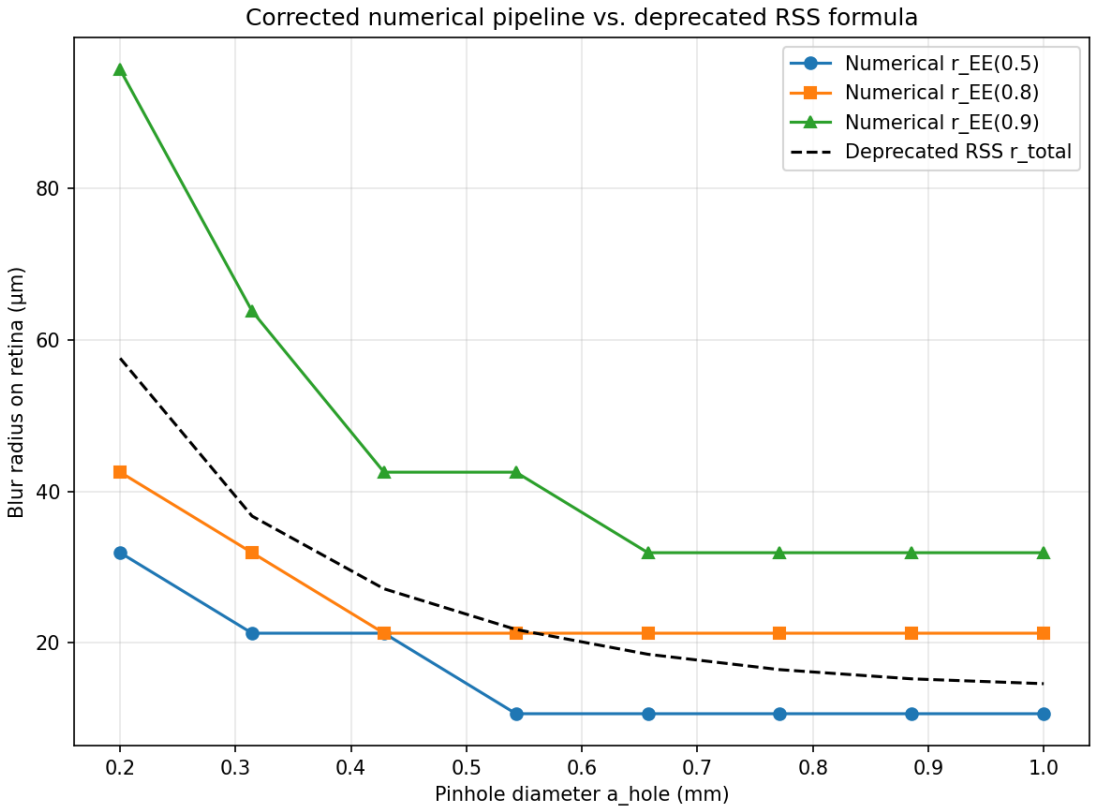


Figure 5: Single-hole sweep: corrected numerical $r_{EE}(P)$ curves vs. the deprecated RSS estimate, across pinhole diameter. Numerical and RSS optima are shown to diverge.

least two myopia severities (Chapter 3's parametrization) — a single number without a stated P and M misrepresents the result.

Pareto optimisation (bridges Part II and Part III): once $a_{hole}(P)$ is fixed, the same $r_{EE}(P)$ value becomes an input to Chapter 12's no-ghost constraint and Chapter 11's transmission target — worked through explicitly in Part VIII.

PART III — Appropriate Pattern Discovery

This answers: how should the holes be arranged?

Chapter 10. Packing Geometry

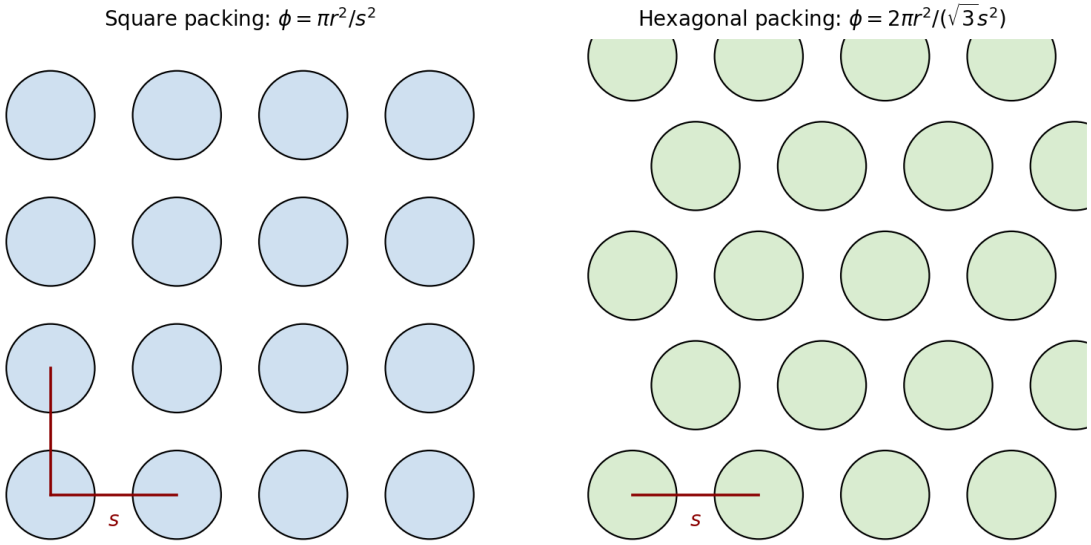


Figure 6: Square vs. hexagonal packing of equal-radius holes at equal center-to-center spacing s

10.1 — Square packing open-area fraction

$$\phi_{square} = \frac{\pi r_{hole}^2}{s^2}$$

Derivation: one hole of area πr_{hole}^2 per unit cell of area s^2 in a square lattice.

10.2 — Hexagonal packing open-area fraction

$$\phi_{hex} = \frac{2\pi r_{hole}^2}{\sqrt{3}\, s^2}$$

Derivation: a hexagonal (triangular) lattice's unit cell is a rhombus of area $s^2 \sin(60^\circ) = s^2\sqrt{3}/2$, containing one hole (by the standard counting argument: each lattice point contributes one full hole to one unit cell in a triangular tiling) — giving $\phi_{hex} = \pi r_{hole}^2/(s^2\sqrt{3}/2) = 2\pi r_{hole}^2/(\sqrt{3}s^2)$. *Comparison:* $\phi_{hex}/\phi_{square} = 2/\sqrt{3} \approx 1.155$ — hexagonal packing gives $\sim 15.5\%$ higher transmission at *equal* spacing s. Since the no-ghost constraint (Chapter 12) caps the *maximum* allowable s and manufacturing (Chapter 14) sets a *minimum* allowable s, hexagonal packing lets you reach a higher transmission target within that same allowed window — a genuine, usable engineering conclusion.

Other patterns (radial, Fermat/golden-angle spiral, Voronoi, random): reduce the perceptual salience of ghosting by breaking the periodicity that makes a regular grid's ghost images themselves periodic and hence more noticeable — a real, plausible qualitative effect, but with **no closed-form open-area or ghosting formula**; if pursued, quantify via controlled photographic/subjective double-vision testing rather than asserting the benefit analytically.

Chapter 11. Brightness

11.1 — Open-area transmission fraction

$$T = \frac{N \cdot (\pi a_{hole}^2/4)}{A_{total}} = \phi_{pattern}$$

11.2 — Transmitted intensity

$$I_{out} = T \cdot I_{in}$$

Optical density (compressed definition): $OD = -\log_{10} T$. Typical commercial sunglasses target $T \approx 15$–25% ($OD \approx 0.6$–0.8) — a reasonable concrete design target to state before solving for N.

Chapter 12. Ghost Images

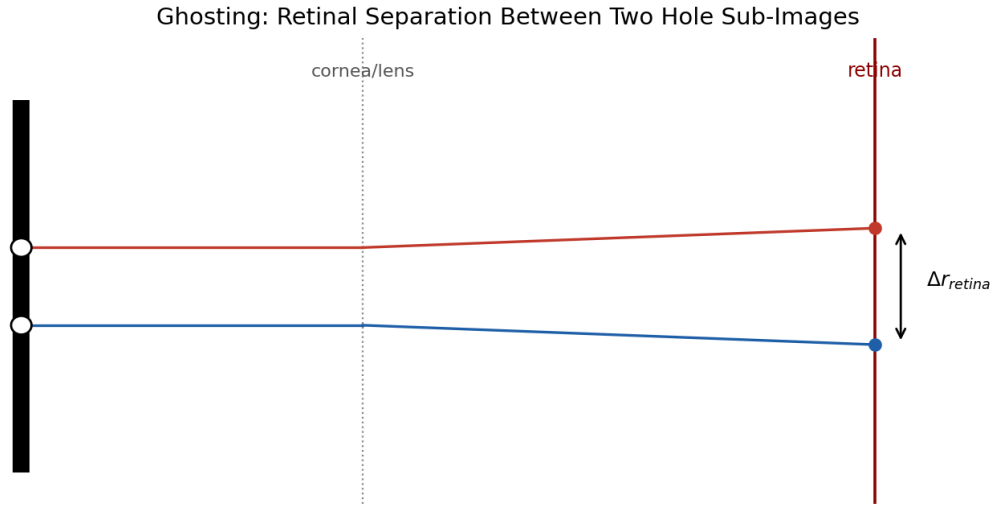


Figure 7: Ghosting geometry: two pinholes separated by s produce two retinal sub-images separated by Δr_{retina}

12.1 — Retinal image shift between two holes

$$\Delta r_{retina} \approx s \cdot \frac{v_r}{d_{gap}}$$

Derivation: similar triangles — a hole offset by s from the axis relays the same distant object as an axial hole would, but shifted; over the gap d_{gap}, this projects to a shift s at the cornea, then magnifies by v_r/d_{gap} through the eye's imaging onto the retina (small-angle approximation, consistent with the paraxial treatment throughout).

12.2 — No-ghost condition

$$s < r_{EE}(P) \cdot \frac{d_{gap}}{v_r}$$

Use: combine with Chapter 10/11's transmission formulas to bound the achievable (N, s) region for a stated transmission target — solved jointly in Part VIII.

Interference/coherence assumption (stated explicitly, not silently assumed): for spatially incoherent extended-source viewing (ordinary daylight scenes — the actual use case), cross-hole interference terms average out and do not need to be modeled; this licenses treating each hole's contribution independently rather than as a coherent multi-aperture interferometer, which would be the (different, and not applicable here) correct treatment only for a coherent point source such as a laser.

Chapter 13. Field of View

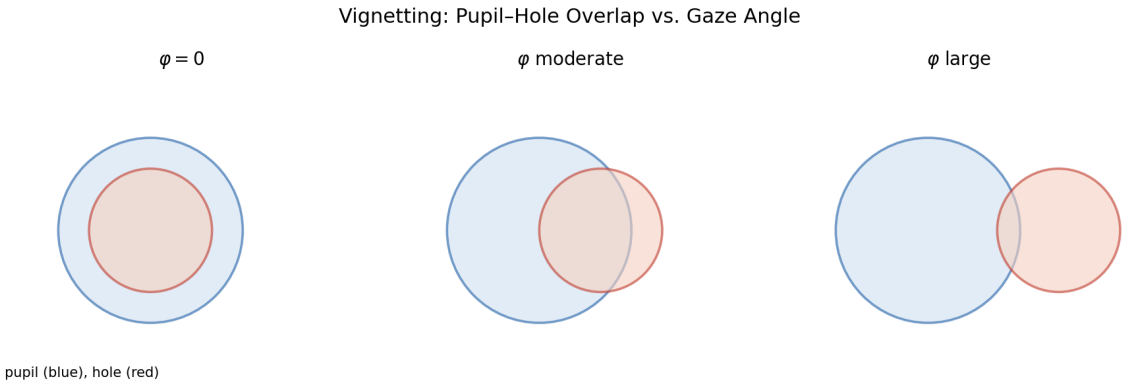


Figure 8: Vignetting: pupil–hole overlap area shrinks as gaze angle increases

13.1 — Pupil lateral displacement under eye rotation

$$\Delta(\varphi) \approx R_{rot} \sin\varphi$$

Derivation: the eye rotates approximately about a single fixed center; the pupil, offset from that center by $R_{rot} \approx 13$–14 mm (physiological value), sweeps laterally by simple circular-arc projection as the eye rotates by φ from the primary gaze position. *Validity:* restrict quantitative use to normal gaze range ($|\varphi| \lesssim 20$–$30°$), consistent with the paraxial/small-angle assumptions used throughout this handbook.

13.2 — Overlap area of two circular apertures (pupil radius R_p, hole radius R_h, center separation Δ)

Derivation. Each circle's boundary intersects the other at two points; the overlap ("lens"/vesica shape) decomposes into two circular segments. For circle 1 (radius R_p), the half-angle subtended at its center by the chord connecting the intersection points is $\alpha_1 = \cos^{-1}\left(\frac{\Delta^2+R_p^2-R_h^2}{2\Delta R_p}\right)$ (law of cosines on the triangle formed by both centers and one intersection point); the circular segment area is $R_p^2(\alpha_1 - \sin\alpha_1\cos\alpha_1) = R_p^2\alpha_1 - \frac{1}{2}R_p^2\sin(2\alpha_1)$. Summing both circles' segment areas and simplifying the shared triangle term (via Heron's-formula-style factoring) gives the closed form:

$$A_{overlap}(\Delta) = R_p^2\cos^{-1}\left(\frac{\Delta^2+R_p^2-R_h^2}{2\Delta R_p}\right) + R_h^2\cos^{-1}\left(\frac{\Delta^2+R_h^2-R_p^2}{2\Delta R_h}\right) - \frac{1}{2}\sqrt{(-\Delta+R_p+R_h)(\Delta+R_p-R_h)(\Delta-$$

valid for $|R_p - R_h| < \Delta < R_p + R_h$; $A = \pi\min(R_p, R_h)^2$ for $\Delta \le |R_p - R_h|$ (one circle fully inside the other); $A = 0$ for $\Delta \ge R_p + R_h$.

13.3 — Acceptance angle. Define φ_{50} as the gaze angle where $A_{overlap}(\varphi)/A_{overlap}(0) = 0.5$, solved numerically from 13.1+13.2. For the *array* (not one hole), sum the per-hole overlap area across all N hole positions at each φ — peripheral holes extend the effective field of view beyond any single hole's own acceptance angle, since a gaze angle that vignettes a central hole may still transmit well through an off-axis one.

Retinal curvature caveat: the flat-retina assumption (Chapter 3) is weakest here — state this explicitly rather than implying this vignetting model alone fully characterizes peripheral image quality.

Chapter 14. Manufacturing

- **Laser cutting:** minimum clean hole diameter $\approx$ 100–150 μm in thin polymer film; heat-affected-zone (HAZ) rim can reduce effective clear aperture by tens of microns — always measure actual post-cut diameter under magnification, never trust the nominal cut value, especially near this project's optimum ($\sim$ 0.3–0.5 mm), where a 20–30 µm HAZ is a several-percent systematic error.
- **Mechanical punching:** cleaner edges, no HAZ, but tooling-limited to $\gtrsim$ 300 μm typically; better dimensional repeatability across many holes, less flexible (new tooling per diameter).
- **Hole circularity tolerance:** propagates directly into Formula 6.1's symmetry assumption — even a few percent ellipticity measurably distorts the Airy pattern; measure real circularity before comparing model to experiment.
- **Surface roughness (edge quality):** a rough or burred hole edge acts as a random phase/amplitude perturbation on the aperture function, scattering some energy outside the ideal Airy pattern — qualitatively reduces measured MTF relative to the ideal simulation; not modeled quantitatively here, flagged as a expected source of model-vs-experiment discrepancy.

PART IV — Lens Material and Thickness

What should the opaque film itself be made from?

Although the official IYPT statement does not explicitly require material selection, it is a legitimate engineering extension since the substrate affects real optical performance (Chapter 16's thickness effect) — treated here at the depth the physics actually demands, not inflated further, per this handbook's stated scope.

Chapter 15. Material Properties

Material	Typical role	Laser-cut edge	Practical min. hole Ø	Notes
Black PET/polyester film	Pinhole substrate	Good, some HAZ	$\sim 150\ \mu$m	Thin, flexible, cheap, laminates easily
Anodized/black aluminum foil	High-opacity thin layer	Very good	~ 100–$150\ \mu$m	Best edge sharpness for small holes
Acrylic (PMMA), opaque-dyed	Rigid frame-mounted film	Moderate HAZ, discoloration risk	~ 200–$300\ \mu$m	Thicker stock triggers Chapter 16's FOV constraint
Polycarbonate	Frame/lens body (not the perforated layer itself)	N/A	—	Relevant to surrounding structure, impact resistance
ABS	3D-printed frame components	N/A (not typically laser-cut for holes this small)	—	Structural only, not the optical film
Stainless steel shim stock	Durable, precise array	Excellent	~ 150–$300\ \mu$m	Heavier, very repeatable manufacturing
Kapton (polyimide film)	High-temperature-tolerant thin film	Good	~ 100–$150\ \mu$m	Relevant only if thermal exposure is a real design concern (Chapter 18)

Chapter 16. Thickness

14.1/16.1 — Tunnel / acceptance-angle model

$$\mathrm{FOV}_{hole} = 2\tan^{-1}\left(\frac{a_{hole}}{2t}\right)$$

Derivation: elementary trigonometry on a cylindrical hole of diameter a_{hole} and depth (film thickness) t — a ray only clears the hole without touching its side wall if its angle from the axis is within

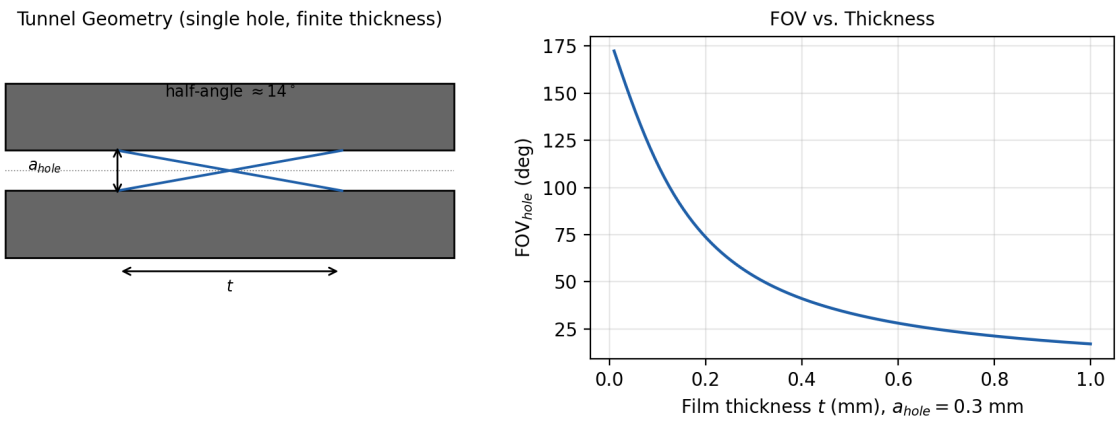


Figure 9: Left: tunnel geometry of a single hole with finite film thickness, defining the acceptance half-angle. Right: field of view vs. thickness for a fixed 0.3 mm hole diameter.

$\pm\tan^{-1}(a_{hole}/2t)$. *Use:* only a binding constraint if t is not negligible relative to a_{hole}. Worked example: $a_{hole} = 0.3$ mm, $t = 0.5$ mm (thick dyed acrylic) $\Rightarrow$ FOV$_{hole} \approx 31°$, a real constraint stacking with Chapter 13's rotational vignetting; the same hole in $t = 50$ μm PET $\Rightarrow\approx 160°$, negligible. **Design conclusion:** choose thin film specifically to keep this term out of the way — state this as a deliberate decision informed by the formula, not an incidental material choice.

Aspect ratio (t/a_{hole}) is the compact way to report this constraint across designs — values below ~ 0.5 keep FOV$_{hole}$ above $\sim 90°$, effectively removing this constraint from the design problem.

Diffraction from finite-depth apertures / waveguide effects: only relevant when t approaches a_{hole} closely enough that the hole behaves as a short waveguide rather than a thin screen — for the thin-film designs recommended above (aspect ratio $\ll 1$), this regime is not reached and is not modeled further here; flagged as a real but second-order effect specific to thick-substrate designs only.

Chapter 17. Internal Reflections

Fresnel reflection at a hole's cylindrical wall matters only in the same thick-substrate regime as the waveguide note above; for thin-film substrates chosen via Chapter 16's formula, negligible energy grazes the wall at all, so a per-bounce reflectance-coefficient treatment is not needed. If a thick, machined substrate is used instead, revisit this with the standard Fresnel reflectance formula $R = \left(\frac{n_1-n_2}{n_1+n_2}\right)^2$ (normal incidence) applied per internal bounce, with black-anodized or matte-black coatings reducing R substantially versus bare metal or polymer.

Chapter 18. Thermal Behaviour

At normal outdoor eyewear operating temperatures (roughly $-10\,^\circ$C to $50\,^\circ$C) and this project's sub-millimeter feature sizes, thermal expansion changes hole diameter by:

$$\Delta a_{hole} = a_{hole} \cdot \alpha \cdot \Delta T$$

Typical polymer/metal linear expansion coefficients $\alpha \sim 10^{-5}$–10^{-4} /°C, applied to $a_{hole} \sim 0.3$ mm over a $60\,^\circ$C range, give Δa_{hole} on the order of 10^{-4}–10^{-3} μm to a few μm at worst — below

this project's manufacturing tolerance (Chapter 14) and below the sensitivity floor established in Chapter 9's flat-minimum check. Stating this magnitude and why it is dropped is the correct level of rigor here; a chapter of expansion-coefficient tables would not change any design decision in this project.

PART V — Simulation

Implemented as `pinhole_sim.py` (accompanying this handbook): circular aperture generation, Angular Spectrum Method propagation (Formula 7.1) with evanescent-mode cutoff, lens phase transformation (8.1), the corrected two-stage pipeline (Chapter 8), numerical radial encircled-energy extraction (6.3), and a validation routine confirming convergence to the closed-form Airy result with increasing grid resolution (Chapter 8's figure). GPU acceleration was not required at this problem's grid sizes — CPU FFT runtimes of tens of seconds per propagation at $N = 4096$ were acceptable for an order-10-point sweep; only worth pursuing for full 2D multi-hole array simulations rather than the validated single-hole, radially-symmetric case.

Sampling was the actual limiting factor encountered (Chapter 7) — resolved by direct convergence testing, not by an a priori formula. **Validation** is documented in Chapter 8's figure; treat any reported number from this pipeline as provisional until that same convergence check has been re-run at the specific (a_{hole}, d_{gap}, v_r) combination being reported, since convergence behavior is parameter-dependent.

PART VI — Experimental Verification

Artificial eye: converging lens ($f \approx 15$–20 mm) with sensor/screen mounted behind the paraxial focal plane by a controlled offset, simulating a stated myopia diopter (Chapter 3's parametrization) with a fixed, repeatable accommodation state (unlike a biological eye).

USAF 1951 resolution target imaged through interchangeable hole films at the parameters identified in Parts II–III.

MTF extraction via slanted-edge or line-target analysis on the captured images (Chapter 6), compared directly against the simulated pipeline's own MTF.

Lux meter/photodiode measurement of transmitted intensity, calibrated against Formulas 11.1–11.2.

Contrast and resolution reported jointly (not resolution alone) since Chapter 6 established that a null-radius or single blur-radius metric misrepresents energy distribution.

Human testing, if pursued: acuity charts (Snellen/logMAR) viewed through candidate designs by volunteer subjects, cross-referenced against the artificial-eye rig's MTF prediction for the same design — a genuine three-way check (theory, artificial-eye measurement, human-subject report).

Error analysis: report measured hole diameter/circularity (Chapter 14) as the dominant expected source of model-vs-experiment discrepancy, alongside grid-convergence uncertainty from Part V and any polychromatic-averaging approximation from Chapter 6.

PART VII — Optimisation (Final Design)

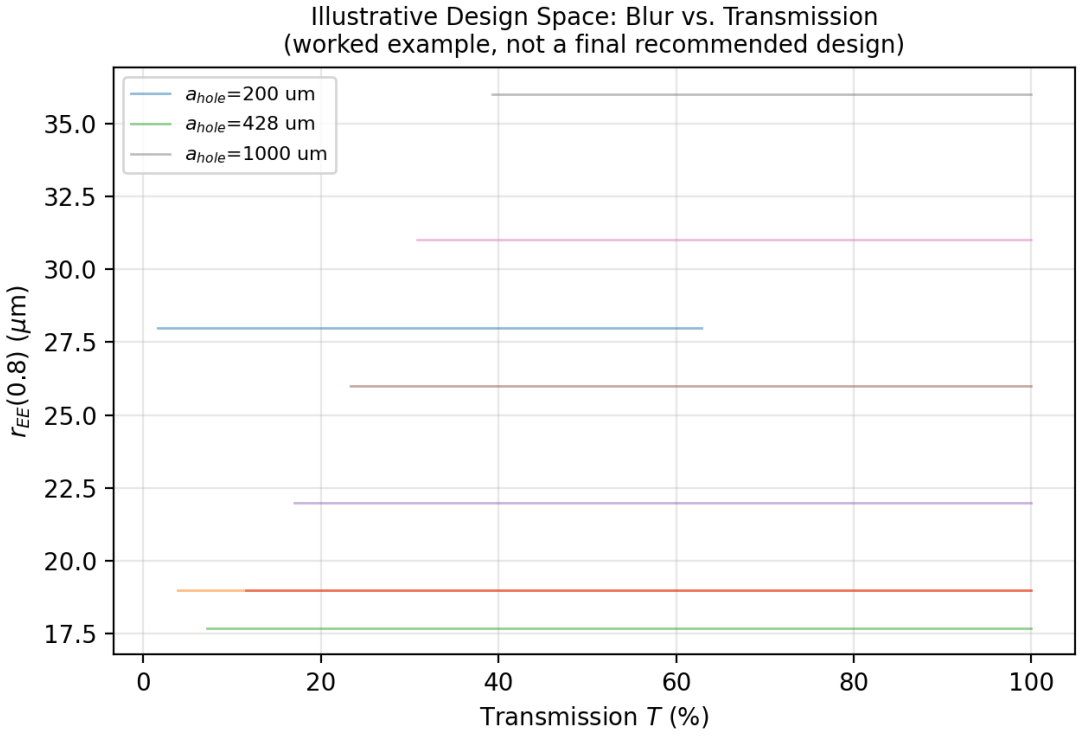


Figure 10: Illustrative worked example combining this project's actual $r_{EE}(0.8)$ sweep values with the transmission formula (11.1) across a range of hole counts N — demonstrates the method, not a final recommended design

Multi-objective optimisation, final sequence:

1. Fix $a_{hole}(P)$ from Chapter 9's numerical sweep, for a chosen P and myopia severity M.
2. Choose hexagonal packing (Chapter 10) for its transmission advantage at equal spacing.
3. Solve Formulas 11.1 and 12.2 jointly for N and s, given a stated transmission target T and the no-ghost bound.
4. Check the resulting design against Chapter 13's acceptance angle and Chapter 14's manufacturing tolerance.
5. Report the full Pareto front of $r_{EE}(P)$ vs. T across candidate designs meeting all constraints, with an explicit, justified tradeoff statement — not a single unqualified "optimal design."

Sensitivity: report how the final recommendation shifts under ± 1 D myopia-severity uncertainty and under the manufacturing tolerance from Chapter 14 — a design that is only optimal at one exact, unmeasurable severity is not a usable recommendation.

Appendix A — Symbol Table

Symbol	Meaning	Unit
a_{hole}	physical pinhole diameter	m
a_{eff}	effective spot diameter after gap propagation (deprecated)	m
a_{stop}	effective aperture stop after pupil clipping	m
d_{gap}	pinhole film to cornea distance	m
v_r	lens-to-retina distance	m
v_f	lens-to-actual-focus distance	m
u	object distance	m
f	eye's pathological focal length	m
λ	wavelength	m
N_F	Fresnel number	—
r_{geo}	geometric blur radius	m
r_{diff}	diffraction first-null radius	m
$r_{EE}(P)$	radius enclosing energy fraction P	m
$EE(x)$	encircled energy function	—
N	hole count	—
s	hole pitch	m
T	open-area transmission fraction	—
M	myopia magnitude	D (m^{-1})
φ	eye gaze rotation angle	rad/deg
R_{rot}	center-of-rotation to pupil-plane distance	m
t	film substrate thickness	m
R_p, R_h	pupil, hole radii (Chapter 13)	m

Appendix B — Formula Index

5.1 Thin lens · 5.2 Geometric blur radius · 6.1 Airy intensity · 6.2 First-null radius · 6.3 Encircled energy · 6.4 Fresnel number · 7.1 Angular Spectrum Method · 7.2 Quadrature effective-aperture (deprecated) · 7.3 On-axis Fresnel intensity · 8.1 Lens phase transformation · 9 Optimum aperture (closed-form and numerical) · 10.1 Square packing fraction · 10.2 Hexagonal packing fraction · 11.1 Open-area transmission · 11.2 Transmitted intensity · 12.1 Retinal image shift · 12.2 No-ghost condition · 13.1 Pupil lateral displacement · 13.2 Circular-aperture overlap area · 13.3 Acceptance-angle definition · 14.1/16.1 Tunnel/FOV thickness model · 18.1 Thermal expansion of hole diameter.

Appendix C — Simulation Pseudocode

```
function run_pipeline(a_hole, d_gap, v_r, f, wavelength, D_pupil, N, L):
    dx = L / N
    U0 = circular_aperture(N, dx, a_hole)
```

```
U1 = ASM_propagate(U0, dx, wavelength, d_gap)      # Formula 7.1
U2 = U1 * circular_aperture(N, dx, D_pupil)        # pupil clip
U3 = U2 * lens_phase(N, dx, wavelength, f)         # Formula 8.1
U4 = ASM_propagate(U3, dx, wavelength, v_r)        # Formula 7.1 again (corrected step)
I_retina = |U4|^2
r_bins, EE = radial_encircled_energy(I_retina, dx) # Formula 6.3, numerical
return I_retina, r_bins, EE
```

Complexity: each ASM propagation is dominated by a 2D FFT, $O(N^2 \log N)$; a full single-hole sweep of k aperture values costs $O(k \cdot N^2 \log N)$. This project's validated run used $N = 4096$, costing on the order of tens of seconds per propagation on CPU — acceptable for the $k \sim 8$–12 point sweeps used throughout, not yet optimized (e.g. via GPU) since not required at this scale.

Full working implementation (validation routine, single-hole sweep, and this pseudocode's real counterpart) accompanies this handbook as `pinhole_sim.py`.